

US Patent & Trademark Office

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United States Patent Application Publication

20250279567

Kind Code

A1

Publication Date

September 04, 2025

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RESONANT FILTER

Abstract

A resonant filter includes a casing, metal resonant sheets, an input port and an output port. The casing has an accommodating cavity and a first inner surface and a second inner surface arranged opposite to each other, and is provided with a first through hole and a second through hole communicating with the accommodating cavity. The metal resonant sheets are located in the accommodating cavity and are disposed on the first inner surface and the second inner surface. The metal resonant sheets are substantially located in a same plane and are distributed oppositely. Coupling occurs between the metal resonant sheets to form a signal connection. The input port is engaged with the first through hole and is connected to one metal resonant sheet. The output port is engaged with the second through hole and is connected to another metal resonant sheet.

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Appl. No.: 19/032970

Filed: January 21, 2025

Foreign Application Priority Data

CN 202410242503.3

Mar. 04, 2024

Publication Classification

Int. Cl.: H01P7/06 (20060101)

U.S. Cl.:

Background/Summary

CROSS REFERENCE TO RELATED PRESENT DISCLOSURE

[0001] This application claims the priority benefit of Chinese Patent Application Serial Number 202410242503.3, filed on Mar. 4, 2024, the full disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of filters, in particular to a resonant filter.

RELATED ART

[0003] A filter is a frequency selective component used to filter out signals in useless frequency bands (i.e., attenuate signals outside the passband) and retain signals in useful frequency bands (i.e., allow signals within the passband to pass).

[0004] The existing resonant filter mainly comprises a housing provided with a cavity and an opening, a cover plate covering the opening, and a resonant member disposed in the cavity. The resonant member is generally of multi-row design and comprises a bending member, a plurality of metal resonant sheets, or a combination thereof. The resonant member of multi-row design occupies a large space in the cavity, which is not conducive to the miniaturization design of the resonant filter. There are problems of high processing difficulty and low dimensional accuracy in the resonant member including the bending member.

[0005] Therefore, it is urgent to provide a resonant filter to solve the above problems.

SUMMARY

[0006] The present disclosure provides a resonant filter, which includes a casing, a plurality of metal resonant sheets, an input port and an output port, wherein the casing has an accommodating cavity and a first inner surface and a second inner surface arranged opposite to each other, and is provided with a first through hole and a second through hole communicating with the accommodating cavity; the plurality of metal resonant sheets are located in the accommodating cavity and disposed on the first inner surface and the second inner surface, the plurality of metal resonant sheets are substantially located in the same plane and are distributed oppositely, and coupling occurs between the plurality of metal resonant sheets to form a signal connection; the input port is engaged with the first through hole and connected to one of the plurality of metal resonant sheets; the output port is engaged with the second through hole and connected to another of the plurality of metal resonant sheets.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Accompanying drawings described herein are intended to provide a further understanding of the present disclosure and form a part of the present disclosure, and exemplary embodiments of the present disclosure and descriptions thereof are intended to explain the present disclosure but are not intended to unduly limit the present disclosure. In the drawings:

[0008] FIG. 1 is a three-dimensional diagram of a resonant filter according to an embodiment of the present disclosure;

[0009] FIG. 2 is an exploded schematic diagram of an embodiment of the resonant filter of FIG. 1;

[0010] FIG. 3 is a cross-sectional schematic diagram of the resonant filter of FIG. 2;

[0011] FIG. 4 is a three-dimensional diagram of a resonant filter according to another embodiment of the present disclosure;

[0012] FIG. 5 is an exploded schematic diagram of an embodiment of the resonant filter of FIG. 4;

[0013] FIG. 6 is a cross-sectional schematic diagram of the resonant filter of FIG. 5;

[0014] FIG. 7 is an exploded schematic diagram of another embodiment of the resonant filter of FIG. 1;

[0015] FIG. 8 is a cross-sectional schematic diagram of the resonant filter of FIG. 7;

[0016] FIG. 9 is a diagram illustrating passband insertion loss curves of the resonant filter of FIG. 3 and the resonant filter of FIG. 8;

[0017] FIG. 10 is a three-dimensional diagram of a resonant filter according to still another embodiment of the present disclosure;

[0018] FIG. 11 is an exploded schematic diagram of an embodiment of the resonant filter of FIG. 10;

[0019] FIG. 12 is a cross-sectional schematic diagram of the resonant filter of FIG. 11;

[0020] FIG. 13 is an exploded schematic diagram of another embodiment of the resonant filter of FIG. 10; and

[0021] FIG. 14 is a cross-sectional schematic diagram of the resonant filter of FIG. 13.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0022] The embodiments of the present disclosure will be described below in conjunction with the relevant drawings. In the figures, the same reference numbers refer to the same or similar components or method flows.

[0023] It must be understood that the words “including”, “comprising” and the like used in this specification are used to indicate the existence of specific technical features, values, method steps, work processes, elements and/or components. However, it does not exclude that more technical features, values, method steps, work processes, elements, components, or any combination of the above can be added.

[0024] It must be understood that when an element is described as being “connected” or “coupled” to another element, it may be directly connected or coupled to another element, and intermediate elements therebetween may be present. In contrast, when an element is described as “directly connected” or “directly coupled” to another element, there is no intervening element therebetween.

[0025] In addition, although the terms such as “first”, “second”, and the like may be used herein to describe different elements, the terms are used only to distinguish the elements or operations described in the same technical terms.

[0026] The embodiments of the present disclosure provide a resonant filter. In some embodiments, the problem that it is not conducive to the miniaturization design of the existing resonant filter due to the resonant member of multi-row design can be solved. In some embodiments, the problems of high processing difficulty and low dimensional accuracy due to the bending member of the resonant member of the existing resonant filter can be solved.

[0027] Please refer to FIG. 1 to FIG. 3, FIG. 1 is a three-dimensional diagram of a resonant filter according to an embodiment of the present disclosure, FIG. 2 is an exploded schematic diagram of an embodiment of the resonant filter of FIG. 1, and FIG. 3 is a cross-sectional schematic diagram of the resonant filter of FIG. 2. As shown in FIG. 1 to FIG. 3, a resonant filter 100 is suitable for frequency bands above 5.5 GHz (Gigahertz), and comprises a casing 110, a plurality of metal resonant sheets 120, an input port 130, and an output port 140. The casing 110 has an accommodating cavity 111 and a first inner surface 112 and a second inner surface 113 arranged opposite to each other, and is provided with a first through hole 114 and a second through hole 115 communicating with the accommodating cavity 111. The plurality of metal resonant sheets 120 are located in the accommodating cavity 111 and disposed on the first inner surface 112 and the second inner surface 113 (that is., the plurality of metal resonant sheets 120 are distributed in two rows), the plurality of metal resonant sheets 120 are substantially located in the same plane and are

distributed oppositely, and coupling occurs between the plurality of metal resonant sheets **120** to form a signal connection. The input port **130** is engaged with the first through hole **114** and is connected to one of the plurality of metal resonant sheets **120**. The output port **140** is engaged with the second through hole **115** and is connected to another metal resonant sheet **120** of the plurality of metal resonant sheets **120**.

[0028] Each of the plurality of metal resonant sheets **120** may be, but is not limited to, a sheet material with a metal surface or a metal sheet. Therefore, the plurality of metal resonant sheets **120** do not need to be bent, and have the advantage of simple processing. Since the plurality of metal resonant sheets **120** are substantially located in the same plane and are distributed oppositely, the flat design of the resonant filter **100** is realized (that is, the length of the resonant filter **100** along the first direction F is reduced, and the first direction F is perpendicular to a second direction; in the embodiment, the first direction F is the thickness direction of the resonant filter **100**, and the second direction is the arrangement direction C of the plurality of metal resonant sheets **120**), which is conducive to the miniaturization of the resonant filter **100**. In addition, the coupling occurs between the two adjacent metal resonant sheets **120** forms the signal connection, and the design of bending member is not required, so there is no problem of high processing difficulty and low dimensional accuracy. Besides, the thickness of the metal resonant sheet **120** may be, but not limited to, 0.5 millimeters (mm) to 3 mm; the thicknesses of the plurality of metal resonant sheets **120** are usually the same, if not, the difference in thickness may be, but not limited to, less than 50%; the thickness of the metal resonant sheet **120** is related to the size and target requirements of the resonant filter **100**. Furthermore, the shape of the metal resonant sheet **120** may be designed according to its frequency and the coupling requirements with the adjacent metal resonant sheets **120**, while ensuring that the size of the metal resonant sheet **120** is as small as possible.

Furthermore, the plurality of metal resonant sheets **120** may be fixed on the first inner surface **112** or the second inner surface **113** by laser welding, solder paste welding or brazing.

[0029] In one embodiment, the material of the plurality of metal resonant sheets **120** may be, but not limited to, iron or iron alloy, and the processing method of the metal resonant sheets **120** may be, but not limited to, laser cutting, wire cutting, or sheet metal stamping.

[0030] In one embodiment, the plurality of metal resonant sheets **120** may be arranged alternately on the first inner surface **112** and the second inner surface **113**.

[0031] In one embodiment, each of the plurality of metal resonant sheets **120** may comprise an upright section **121**, an extension section **123** extending from one end of the upright section **121**, and a coupling section **125** extending from one end of the extension section **123** away from the upright section **121**. One end of the upright section **121** away from the extension section **123** is fixed on the first inner surface **112** or the second inner surface **113**. The coupling section **125** of each of the plurality of metal resonant sheets **120** is close to the coupling section **125** of the adjacent metal resonant sheet **120**. The two adjacent metal resonant sheets **120** may be two metal resonant sheets **120** arranged opposite to each other, or two metal resonant sheets **120** arranged along the arrangement direction C.

[0032] In one embodiment, the plurality of metal resonant sheets **120** are not physically connected to each other. In one example, the plurality of metal resonant sheets **120** fixed on the first inner surface **112** are arranged at intervals, and the plurality of metal resonant sheets **120** fixed on the second inner surface **113** are arranged at intervals; the distance between two adjacent metal resonant sheets **120** fixed on the first inner surface **112** and the second inner surface **113** is related to the size and target requirements of the resonant filter **100**.

[0033] In one embodiment, the casing **110** is further provided with a plurality of partition ribs **116**, and the plurality of partition ribs **116** divide the accommodating cavity **111** into a plurality of resonant cavities **117** arranged in two rows, the plurality of metal resonant sheets **120** are disposed in the plurality of resonant cavities **117**, and the two adjacent resonant cavities **117** communicate with each other. Specifically, the adjacent upper and lower rows of resonant cavities **117**

communicate with each other, and the plurality of metal resonant sheets **120** are disposed in the plurality of resonant cavities **117**, so that coupling occurs between the two metal resonant sheets **120** arranged oppositely to form the signal connection. One resonant cavity **117** may be provided with one or more metal resonant sheets **120**, and the number of resonant cavities **117**, the number of metal resonant sheets **120** and the corresponding arrangement relationship between the resonant cavities **117** and the metal resonant plates **120** can be designed according to actual needs.

[0034] In one embodiment, the casing **110** may comprise a bottom plate **118**, a housing **119** and a cover plate **127**, the housing **119** is provided with a first opening **1191** and a second opening **1192** arranged opposite to each other, the bottom plate **118** is disposed at the first opening **1191**, and the cover plate **127** is disposed at the second opening **1192**. The bottom plate **118**, the housing **119** and the cover plate **127** are surrounded to form the accommodating cavity **111**, the bottom plate **118** has the first inner surface **112**, and the cover plate **127** has the second inner surface **113**. The material of the bottom plate **118**, the housing **119** and the cover plate **127** may be, but not limited to, aluminum or aluminum alloy, and the processing method of the housing **119** may be, but not limited to, computer numerical control (CNC) process, die casting, or extrusion molding.

[0035] In one embodiment, the first through hole **114** and the second through hole **115** are arranged on the bottom plate **118** or the cover plate **127**. The input port **130** comprises a first support seat **132**, the metal resonant sheet **120** connected to the input port **130** passes through the first support seat **132** through a first tap plate **120a** and then extends out of the casing **110**, and the first tap plate **120a** and the metal resonance sheet **120** connected to the input port **130** are integrally formed. The output port **140** comprises a second support seat **142**, the metal resonant sheet **120** connected to the output port **140** passes through the second support seat **142** through a second tap plate **120b** and then extends out of the casing **110**, and the second tap plate **120b** and the metal resonant sheet **120** connected to the output port **140** are integrally formed. The material of the first support seat **132** and the second support seat **142** may be, but not limited to, engineering plastics, such as polytetrafluoroethylene (PTFE), perfluoroalkoxy alkane (PFA), and polyphenylene sulfide (PPS); the processing method of the first support seat **132** and the second support seat **142** may be, but not limited to, CNC process, injection molding process, or powder sintering process. The first support seat **132** and the second support seat **142** may be engaged with the first through hole **114** and the second through hole **115** by an adhesive. The first tap plate **120a** is configured to feed the signal into the resonant filter **100**. The second tap plate **120b** is configured to feed the signal out of the resonant filter **100**. The number of the first tap plates **120a** and the number of the second tap plates **120b** are related to the number of channels of the resonant filter **100**. In this embodiment, the resonant filter **100** is a single-channel filter, so the number of the first tap plates **120a** and the number of the second tap plates **120b** are both one.

[0036] Please refer to FIG. 4 to FIG. 6, FIG. 4 is a three-dimensional diagram of a resonant filter according to another embodiment of the present disclosure, FIG. 5 is an exploded schematic diagram of an embodiment of the resonant filter of FIG. 4, and FIG. 6 is a cross-sectional schematic diagram of the resonant filter of FIG. 5. As shown in FIG. 4 to FIG. 6, it can be seen that the difference between the embodiment of FIG. 4 to FIG. 6 and the embodiment of FIG. 1 to FIG. 3 lies in the arrangement positions of the first through hole **114** and the second through hole **115** and the design of the input port **130** and the output port **140**. In FIG. 4 to FIG. 6, the first through hole **114** and the second through hole **115** may be arranged on the same side wall of the housing **119**, and the same side wall may be a side wall **1193** of the housing **119** parallel to the arrangement direction C of the metal resonant sheets **120**. The input port **130** may comprise a first port conductor **134** and a first base **136**, one end of the first port conductor **134** is connected to one metal resonant sheet **120**, the first base **136** is engaged with the first through hole **114**, and the other end of the first port conductor **134** passes through the first base **136** and then extends out of the casing **110**. The output port **140** may comprise a second port conductor **144** and a second base **146**, one end of the second port conductor **144** is connected to another metal resonant sheet **120**, the

second base **146** is engaged with the second through hole **115**, and the other end of the second port conductor **144** passes through the second base **146** and then extends out of the casing **110**. The material of the first base **136** and the second base **146** may be, but not limited to, engineering plastics. The processing method of the first base **136** and the second base **146** may be, but not limited to, CNC process, injection molding process, or powder sintering process. The first base **136** and the second base **146** may be engaged with the first through hole **114** and the second through hole **115** by an adhesive. The first port conductor **134** is configured to feed the signal into the resonant filter **100**. The second port conductor **144** is configured to feed the signal out of the resonant filter **100**. The number of the input port **130** and the number of the output port **140** are related to the number of channels of the resonant filter **100**. In this embodiment, the resonant filter **100** is a single-channel filter, so the number of the input ports **130** and the number of the output ports **140** are both one.

[0037] In one embodiment, the first through hole **114** and the second through hole **115** may be arranged on different side walls of the housing **119**, and the different side walls may be two corresponding side walls of the housing **119** parallel to the arrangement direction C of the metal resonant sheets **120**. The input port **130** may comprise a first port conductor **134** and a first base **136**, one end of the first port conductor **134** is connected to one metal resonant sheet **120**, the first base **136** is engaged with the first through hole **114**, and the other end of the first port conductor **134** passes through the first base **136** and then extends out of the casing **110**. The output port **140** may comprise a second port conductor **144** and a second base **146**, one end of the second port conductor **144** is connected to another metal resonant sheet **120**, the second base **146** is engaged with the second through hole **115**, and the other end of the second port conductor **144** passes through the second base **146** and then extends out of the casing **110**. In other words, the input port **130** and the output port **140** can be arranged in a more flexible manner to meet different design requirements.

[0038] Please refer to FIG. 1 to FIG. 3. The plurality of metal resonant sheets **120** comprise a first resonant sheet **122**, a second resonant sheet **124**, a third resonant sheet **126** and a fourth resonant sheet **128** which are sequentially connected in signal connection. The first resonant sheet **122** and the fourth resonant sheet **128** are arranged at intervals on the second inner surface **113**, the second resonant sheet **124** and the third resonant sheet **126** are arranged at intervals on the first inner surface **112**, the first resonant sheet **122** and the fourth resonant sheet **128** are disposed in different resonant cavities **117**, and the second resonant sheet **124** and the third resonant sheet **126** are disposed in the same resonant cavity **117**. The signal transmission path is the first resonant sheet **122**, the second resonant sheet **124**, the third resonant sheet **126** and the fourth resonant sheet **128** in sequence. Each time the signal passes through a metal resonant sheet **120**, the phase of the signal will change by -90° . The inductive coupling occurs between the first resonant sheet **122** and the second resonant sheet **124**, the inductive coupling occurs between the second resonant sheet **124** and the third resonant sheet **126**, and the inductive coupling occurs between the third resonant sheet **126** and the fourth resonant sheet **128**. The inductive coupling is used to enhance the high-end suppression of the resonant filter **100**. The coupling amount between the first resonant sheet **122** and the third resonant sheet **126** is extremely small, and the coupling amount between the second resonant sheet **124** and the fourth resonant sheet **128** is extremely small, so there is no influence on the implementation of the resonant filter **100**.

[0039] In one embodiment, the input port **130** and the output port **140** are connected to the first resonant sheet **122** and the fourth resonant sheet **128** respectively.

[0040] Please refer to FIG. 7 and FIG. 8, FIG. 7 is an exploded schematic diagram of another embodiment of the resonant filter of FIG. 1, and FIG. 8 is a cross-sectional schematic diagram of the resonant filter of FIG. 7. As shown in FIG. 7 and FIG. 8, it can be seen that the difference between the embodiments of FIG. 7 and FIG. 8 and the embodiments of FIG. 1 to FIG. 3 is that the first resonant sheet **122** and the fourth resonant sheet **128** are disposed in different resonant cavities

117 communicating with each other, and the capacitive cross coupling occurs between the first resonant sheet **122** and the fourth resonant sheet **128**.

[0041] Please refer to FIG. 3, FIG. 8 and FIG. 9, and FIG. 9 is a diagram illustrating passband insertion loss curves of the resonant filter of FIG. 3 and the resonant filter of FIG. 8. In FIG. 9, the horizontal axis represents frequency in Megahertz (MHz), the vertical axis represents insertion loss in dB, the solid line is the passband insertion loss curve of the resonant filter **100** (i.e., the resonant filter **100** of FIG. 8) when the first resonant sheet **122** and the fourth resonant sheet **128** are disposed in different resonant cavities **117** communicating with each other, and the dotted line is the passband insertion loss curve of the resonant filter **100** (i.e., the resonant filter **100** of FIG. 3) when the first resonant sheet **122** and the fourth resonant sheet **128** are disposed in respective independent resonant cavities **117**. As shown in FIG. 9, when the first resonant sheet **122** and the fourth resonant sheet **128** are disposed in respective independent resonant cavities **117**, the first resonant sheet **122** and the fourth resonant sheet **128** are blocked by the partition rib **116**, and no cross coupling occurs, so that the out-of-band signal suppression capability of the resonant filter **100** is relatively poor (i.e., slow signal attenuation). When the first resonant sheet **122** and the fourth resonant sheet **128** are disposed in different resonant cavities **117** communicating with each other, the capacitive cross coupling occurs between the first resonant sheet **122** and the fourth resonant sheet **128**, so that the phase of the signal changes by $+90^\circ$, and symmetrical zero points are realized on the left and right sides of the passband insertion loss curve of the resonant filter **100** (that is, a capacitive zero point is located on the left side of the passband to enhance the low-end suppression of the resonant filter **100**, and an inductive zero point is located on the right side of the passband to enhance the high-end suppression of the resonant filter **100**), thereby significantly improving the out-of-band signal suppression capability of the resonant filter **100**.

[0042] In one embodiment, when the first resonant sheet **122** and the fourth resonant sheet **128** are disposed in different resonant cavities **117** communicating with each other, the resonant filter **100** may further comprise a first connecting rib **150**, the first connecting rib **150** and the first resonant sheet **122** are substantially located in the same plane, and the first connecting rib **150** is connected to the first resonant sheet **122** and the fourth resonant sheet **128** (as shown in FIG. 8). The first connecting rib **150**, the first resonant sheet **122** and the fourth resonant sheet **128** may be, but are not limited to, integrally formed, and the first connecting rib **150** is configured to increase the coupling amount of the capacitive cross coupling between the first resonant sheet **122** and the fourth resonant sheet **128**.

[0043] In one embodiment, the resonant filter **100** may further comprise a second connecting rib **160**, the second connecting rib **160** and the second resonant sheet **124** are substantially located in the same plane, and the second connecting rib **160** is connected to the second resonant sheet **124** and the third resonant sheet **126** (as shown in FIG. 8). The second connecting rib **160**, the second resonant sheet **124** and the third resonant sheet **126** may be, but are not limited to, integrally formed, and the second connecting rib **160** is configured to enhance the electromagnetic coupling between the second resonant sheet **124** and the third resonant sheet **126**.

[0044] In one embodiment, when other factors remain unchanged, the smaller the distance between the end of the second resonant sheet **124** away from the first inner surface **112** and the end of the third resonant sheet **126** away from the first inner surface **112** (that is, the smaller the distance between the coupling section **125** of the second resonant sheet **124** and the coupling section **125** of the third resonant sheet **126**), the smaller the distance between the second connecting rib **160** and the first inner surface **112**; the larger the distance between the end of the second resonant sheet **124** away from the first inner surface **112** and the end of the third resonant sheet **126** away from the first inner surface **112** (that is, the larger the distance between the coupling section **125** of the second resonant sheet **124** and the coupling section **125** of the third resonant sheet **126**), the larger the distance between the second connecting rib **160** and the first inner surface **112**. The larger the distance between the second connecting rib **160** and the first inner surface **112**, the stronger the

coupling effect between the second resonant sheet **124** and the third resonant sheet **126**.

[0045] Please refer to FIG. **10** to FIG. **12**, FIG. **10** is a three-dimensional diagram of a resonant filter according to still another embodiment of the present disclosure, FIG. **11** is an exploded schematic diagram of an embodiment of the resonant filter of FIG. **10**, and FIG. **12** is a cross-sectional schematic diagram of the resonant filter of FIG. **11**. As shown in FIG. **10** to FIG. **12**, it can be seen that the first resonant sheet **122**, the second resonant sheet **124**, the third resonant sheet **126** and the fourth resonant sheet **128** may constitute a group of resonant sheets **129**, and the plurality of metal resonant sheets **120** may comprise N groups of resonant sheets **129** and N-1 intermediate resonant sheets **170**, an intermediate resonant sheet **170** is disposed between two adjacent groups of resonant sheets **129**, N is an integer greater than or equal to 2, and each of the N-1 intermediate resonant sheets **170** is disposed on the first inner surface **112** or the second inner surface **113** and is coupled with the fourth resonant sheet **128** and the first resonant sheet **122** adjacent to two sides thereof (that is, an intermediate resonant sheet **170** is coupled with the fourth resonant sheet **128** of the former group of resonant sheets **129** and the first resonant sheet **122** of the latter group of resonant sheets **129**, wherein the former group of resonant sheets **129** and the latter group of resonant sheets **129** are defined by the sequence of signal transmission) to form the signal connection. In this embodiment, N can be but not limited to 2, so the resonant filter **100** comprises two groups of resonant sheets **129** and an intermediate resonant sheet **170**. The intermediate resonant sheet **170** may be disposed on the first inner surface **112**, and the resonant filter **100** can generate two capacitive zero points and two inductive zero points due to the two groups of resonant sheets **129**. It should be noted that the intermediate resonant sheet **170** is not coupled with the third resonant sheet **126** of the former group of resonant sheets **129** and the second resonant sheet **124** of the latter group of resonant sheets **129**, so the intermediate resonant sheet **170** can be disposed on the first inner surface **112** or the second inner surface **113**.

[0046] In one embodiment, the input port **130** and the output port **140** are connected to the first resonant sheet **122** of the first group of resonant sheets **129** and the fourth resonant sheet **128** of the Nth group of resonant sheets **129** respectively. For example, please refer to FIG. **12**, the input port **130** and the output port **140** are connected to the first resonant sheet **122** of the first group of resonant sheets **129** (i.e., the leftmost group of resonant sheets **129** in FIG. **12**) and the fourth resonant sheet **128** of the second group of resonant sheets **129** (i.e., the rightmost group of resonant sheets **129** in FIG. **12**) respectively. Thus, the signal transmission path is the first resonant sheet **122**, the second resonant sheet **124**, the third resonant sheet **126** and the fourth resonant sheet **128** of the first group of resonant sheets **129**, the intermediate resonant sheet **170**, and the first resonant sheet **122**, the second resonant sheet **124**, the third resonant sheet **126** and the fourth resonant sheet **128** of the second group of resonant sheets **129** in sequence.

[0047] In one embodiment, the intermediate resonant sheet **170** disposed on the second inner surface **113** and the fourth resonant sheet **128** and the first resonant sheet **122** adjacent to two sides thereof are disposed in the same resonant cavity **117**.

[0048] Please refer to FIG. **10**, FIG. **13** and FIG. **14**, FIG. **13** is an exploded schematic diagram of another embodiment of the resonant filter of FIG. **10**, and FIG. **14** is a cross-sectional schematic diagram of the resonant filter of FIG. **13**. As shown in FIG. **13** and FIG. **14**, it can be seen that the difference between the embodiment of FIG. **13** and FIG. **14** and the embodiment of FIG. **11** and FIG. **12** is that the intermediate resonant sheet **170** of FIG. **13** and FIG. **14** is disposed on the first inner surface **112**, and the intermediate resonant sheet **170** disposed on the first inner surface **112** and the second resonant sheet **124** and third resonant sheet **126** adjacent thereto (i.e., the third resonant sheet **126** of the former group of resonant sheets **129** and the second resonant sheet **124** of the latter group of resonant sheets **129**) are disposed in different resonant cavities **117**.

[0049] In summary, in the embodiments of the resonant filter of the present disclosure, the plurality of metal resonant sheets do not need to be bent and have the advantage of simple processing; the plurality of metal resonant sheets are substantially located in the same plane and are distributed

oppositely, and the length of the resonant filter along the direction perpendicular to the arrangement direction of the plurality of metal resonant sheets is significantly reduced, which is conducive to the miniaturization development of the resonant filter; coupling occurs between two adjacent metal resonant sheets to form the signal connection, and the design of bending member is not required, so there is no problem of high processing difficulty and low dimensional accuracy. In addition, the resonant filter of the present disclosure can be a single-layer stacking, drawer-type design with a compact structure, and this architecture provides convenience for subsequent expansion of multiplexers. Besides, the resonant filter of the present disclosure is suitable for frequency bands above 5.5 GHz. Furthermore, the input port and the output port of the resonant filter of the present disclosure can be arranged in a more flexible manner to meet different design requirements.

[0050] While the present disclosure is disclosed in the foregoing embodiments, it should be noted that these descriptions are not intended to limit the present disclosure. On the contrary, the present disclosure covers modifications and equivalent arrangements obvious to those skilled in the art. Therefore, the scope of the claims must be interpreted in the broadest manner to comprise all obvious modifications and equivalent arrangements.

Claims

1. A resonant filter, comprising: a casing having an accommodating cavity, a first inner surface and a second inner surface opposite to the first inner surface, and provided with a first through hole and a second through hole communicating with the accommodating cavity; a plurality of metal resonant sheets located in the accommodating cavity and disposed on the first inner surface and the second inner surface, wherein the plurality of metal resonant sheets are substantially located in a same plane and are distributed oppositely, and the plurality of metal resonant sheets are configured to generate coupling with one another to form a signal connection; an input port engaged with the first through hole and connected to one of the plurality of metal resonant sheets; and an output port engaged with the second through hole and connected to another of the plurality of metal resonant sheets.
2. The resonant filter according to claim 1, wherein the casing is further provided with a plurality of partition ribs, the plurality of partition ribs divide the accommodating cavity into a plurality of resonant cavities arranged in two rows, the plurality of metal resonant sheets are disposed in the plurality of resonant cavities, and two adjacent opposite resonant cavities communicate with each other.
3. The resonant filter according to claim 2, wherein the plurality of metal resonant sheets comprise a first resonant sheet, a second resonant sheet, a third resonant sheet and a fourth resonant sheet which are sequentially connected in signal connection, the first resonant sheet and the fourth resonant sheet are arranged at intervals on the second inner surface, the second resonant sheet and the third resonant sheet are arranged at intervals on the first inner surface, the first resonant sheet and the fourth resonant sheet are disposed in different resonant cavities, and the second resonant sheet and the third resonant sheet are disposed in the same resonant cavity.
4. The resonant filter according to claim 3, wherein the input port and the output port are connected to the first resonant sheet and the fourth resonant sheet respectively.
5. The resonant filter according to claim 3, wherein the different resonant cavities with the first resonant sheet and the fourth resonant sheet communicate with each other, and the first resonant sheet and the fourth resonant sheet are configured to generate capacitive cross-coupling with each other.
6. The resonant filter according to claim 5, further comprising a first connecting rib, wherein the first connecting rib and the first resonant sheet are substantially located in the same plane, and the first connecting rib is connected to the first resonant sheet and the fourth resonant sheet.
7. The resonant filter according to claim 3, further comprising a second connecting rib, wherein the

second connecting rib and the second resonant sheet are substantially located in the same plane, and the second connecting rib is connected to the second resonant sheet and the third resonant sheet.

8. The resonant filter according to claim 7, wherein the smaller a distance between an end of the second resonant sheet away from the first inner surface and an end of the third resonant sheet away from the first inner surface, the smaller a distance between the second connecting rib and the first inner surface; the larger the distance between the end of the second resonant sheet away from the first inner surface and the end of the third resonant sheet away from the first inner surface, the larger the distance between the second connecting rib and the first inner surface.

9. The resonant filter according to claim 3, wherein the first resonant sheet, the second resonant sheet, the third resonant sheet and the fourth resonant sheet constitute a group of resonant sheets, the plurality of metal resonant sheets comprise N groups of resonant sheets and N-1 intermediate resonant sheets, an intermediate resonant sheet of the N-1 intermediate resonant sheets is disposed between two adjacent groups of resonant sheets of the N groups of resonant sheets, N is an integer greater than or equal to 2, and each intermediate resonant sheet of the N-1 intermediate resonant sheets is disposed on the first inner surface or the second inner surface and is configured to be coupled with the fourth resonant sheet and the first resonant sheet adjacent to two sides thereof to form a signal connection.

10. The resonant filter according to claim 9, wherein the input port and the output port are connected to the first resonant sheet of a first group of resonant sheets and the fourth resonant sheet of an Nth group of resonant sheets, respectively.

11. The resonant filter according to claim 9, wherein the intermediate resonant sheet disposed on the second inner surface and the fourth resonant sheet and the first resonant sheet adjacent to two sides thereof are disposed in the same resonant cavity.

12. The resonant filter according to claim 9, wherein the intermediate resonant sheet disposed on the first inner surface and the second resonant sheet and the third resonant sheet adjacent to two sides thereof are disposed in different resonant cavities.

13. The resonant filter according to claim 1, wherein the casing comprises a bottom plate, a housing and a cover plate, the housing is provided with a first opening and a second opening arranged opposite to each other, the bottom plate is disposed at the first opening, the cover plate is disposed at the second opening; the bottom plate, the housing and the cover plate are surrounded to form the accommodating cavity; the bottom plate has the first inner surface, and the cover plate has the second inner surface.

14. The resonant filter according to claim 13, wherein the first through hole and the second through hole are arranged on the bottom plate or the cover plate.

15. The resonant filter according to claim 14, wherein the input port comprises a first support seat, the one metal resonant sheet connected to the input port passes through the first support seat through a first tap plate and extends out of the casing, and the first tap plate and the one metal resonant sheet connected to the input port are integrally formed; the output port comprises a second support seat, the another metal resonant sheet connected to the output port passes through the second support seat through a second tap plate and extends out of the casing, and the second tap plate and the another metal resonant sheet connected to the output port are integrally formed.

16. The resonant filter according to claim 1, wherein the first through hole and the second through hole are arranged on the same side wall or different side walls of the casing.

17. The resonant filter according to claim 16, wherein the input port comprises a first port conductor and a first base, one end of the first port conductor is connected to the one metal resonant sheet, the first base is engaged with the first through hole, and another end of the first port conductor passes through the first base and extends out of the housing; the output port comprises a second port conductor and a second base, one end of the second port conductor is connected to the another metal resonant sheet, the second base is engaged with the second through hole, and another

end of the second port conductor passes through the second base and extends out of the housing.

18. The resonant filter according to claim 1, wherein each of the plurality of metal resonant sheets comprises a upright section, an extension section extending from one end of the upright section, and a coupling section extending from one end of the extension section away from the upright section, an end of the upright section away from the extension section is connected on the first inner surface or the second inner surface, and the coupling section of each of the plurality of metal resonant sheets is close to the coupling section of an adjacent metal resonant sheet.

19. The resonant filter according to claim 1, wherein each of the plurality of metal resonant sheets is a sheet with a metal surface or a metal sheet.

20. The resonant filter according to claim 1, wherein the plurality of metal resonant sheets are not physically connected to each other.
